

SAMARTH YADANNAVAR

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LinkedIn

Portfolio

EDUCATION

Indian Institute of Technology, Bombay

10/2020 - 05/2024

Major :- Bachelors in Metallurgical and Material Science - **CPI - 7.56**

Mumbai

Minor :- Data Science and Artificial Intelligence from CMINDS, IIT Bombay

Relevant Coursework :- Data Analysis and Interpretation, Introduction to C/C++ & Python, Machine Learning, Deep Learning, Computer Vision, NLP, Simulation and Optimization, Learning Analytics and Education Data Mining

WORK EXPERIENCE

AI Scientist | Quant.ai, USA

09/2024 - Present

- Engineered a robust Multi-agent framework by preparing and integrating orthogonal lambda partitioning on the top of Dify for significantly reducing LLM hallucinations
- Led the development of a Banking demo with 10 use cases, integrating Postman APIs and MongoDB for streamlined data retrieval and updates, delivering an end-to-end functional solution
- Driving future AI innovation by developing a fully conversational AI agent capable of autonomously generating lambda functions, enhancing its human-like interaction and problem-solving capabilities

INTERSHIPS

AI Research and Platform Intern | Fractal Analytics, Mumbai

04/2024 - 07/2024

- Implemented **Med-Agents**, a chain-of-thought technique for medical reasoning on the NEET PG dataset, boosting **GPT-4o's** accuracy from **77%** to **81.6%** using a zero-shot approach
- Further modified the given technique to resolve inconsistencies and confusion in the final decision making which led to **1.4%** increase in accuracy, amounting to an overall accuracy of **83%**
- Added Direct Preference Optimization (**DPO**) task for **LLava-Med** Multimodal LLM, enabling the model to learn and interpret unknown medical images autonomously

NLP Engineer | GetWork, Gurgaon

01/2024 - 04/2024

- Developed a **Job Matching Agent** using **OpenAI API** to match students' resume content with job descriptions, streamlining the placement process for students from tier 2 and 3 colleges through automated assistance
- Implemented a **RAG** for enabling semantic search on **MongoDB** vector database, which calculates vector embeddings, and similarity scoring using cosine similarity function
- Utilized **LangChain** Framework for improving the speed and accuracy of matching student resumes with job opportunities, resulting in **200ms** user-query response time

Computer Vision Engineer | Ezynest, USA

04/2024 - 07/2024

- Converted CCTV data of parking lots into tailored **YOLOv8** dataset on Roboflow, ensuring accurate annotations
- Fine-tuned** Ultralytics' YOLOv8 model, maximizing the accuracy of detection of empty parking space
- Integrated **real-time** parking availability model with company's website, enabling customers to find and book vacant spots before arriving

Microexpressions and User Engagement | Research Intern | Prof R. Rajendran

12/2023 - 02/2024

- Curated an online learning dataset comprising data from **10** college students, generated **943,520** frames at **60** frames per second (fps) to facilitate in-depth analysis of learner engagement and behaviour patterns
- Achieved **84%** accuracy in detecting emotions such as happiness, sadness, repression, etc using Motion Magnification Technique for analyzing "microexpressions"
- Attained an accuracy of **95%** on affective state engagement and identified a negative correlation of **-0.76** between learner's emotions and engagement by conducting correlation analysis

AI Summer Intern | Fogteams, Bengaluru

04/2022 - 07/2022

- Developed an AI assistant for Slack and Fogteams website using **Rasa**, automating note-taking and minutes summarization for enhancing efficiency of meetings
- Leveraged Rasa's **extractors**, **Spacy NLU** to process employee data and Facebook's Duckling extractor to extract date and time, enabling personalized communication and timely reminders
- Efficiently coordinated meetings through conversational interactions through **Dialog Management** by fine-tuning on over **1000+** examples, ensuring alignment of employee and client schedules

PUBLICATION

Samarth Yadannavar, Ashwin T.S., Manjunath, Ramkumar Rajendran "Multimodal Analysis of Learning-Centered Emotions and Cognitive Processes in Open-Ended Learning Environments", IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE), December 2024

ENTREPRENEURIAL EXPERIENCE

StartupSense 08/2022 - 11/2023

- Co-founded and served as CTO of StartupSense, leading a team to develop AI-driven humanoids aimed at transforming human-robot interaction
- Spearheaded the development of "Sage," an autonomous humanoid capable of real-time interaction and learning from user behaviors using reinforcement learning algorithms
- Acquired a grant of INR 50,000 from IDEAS program conducted by DSSE School of Entrepreneurship at IITB

KEY PROJECTS

Cancer Cell Detection | Prof. Amit Sethi 02/2024 - 05/2024

- Employed Vision Transformer & KimiaNet model for early cancer detection, training on WSI data from 4 organs
- Implemented domain generalization via transfer learning, achieving 0.7 AUC for cross-organ cancer classification
- Explored methods like synthetic data generation, Weighted Cross Entropy & Focal loss to handle class imbalance

Spiking Neural Network | Prof. Udayan Ganguly 08/2023 - 11/2023

- Improved accuracy of Liquid State Machine, a type of Spiking Neural Network on N-MNIST from 95% to 98%
- Extracted receptive field-based features using 18 Gabor Filters, inspired by hierarchical processing of visual cortex
- Enhanced performance by ensembling filters & CNN with LSM made of 3600 neurons without increasing model size

Image Colorization | Season of Code Project 08/2023 - 12/2023

- Developed GAN with U-Net generator and PatchGAN discriminator, achieving a realistic colorization with over 90% visual similarity compared to ground truth
- Engineered custom loss function combining perceptual, adversarial, & L1 losses for realistic colorization outputs
- Leveraged Optuna for hyperparameter tuning, achieving 40% faster training while preserving high-quality output

IoT: Home Automation | Institute Techincal Summer Project 05/2023 - 07/2023

- Designed voice-controlled home automation system using Raspberry Pi & ESP32 to control appliances effortlessly
- Utilized Google Firebase to facilitate real-time data storage, retrieval, & hosting the HTML-based control interface
- Engineered a ChatGPT-based Conversational Chatbot using Speech Recognition & Google TTS for audio processing

POSITION OF RESPONSIBILITY

Head of Data Analytics and Visualization | IIT Bombay 07/2023 - 05/2024

- Performed data analysis using Python and AI algorithms to compare grading statistics pre and post-COVID.
- Prepared an extensive report catering to 500+ professors to recommend enhancements to the curriculum

TECHNICAL SKILLS

Languages and Databases: Python, R, C/C++, HTML/CSS, SQL, no-SQL, MongoDB, PostgreSQL, LATEX, MATLAB

Big Data: S3 Buckets, EBS, Lambda, Azure, GCP, Sagemaker

Tools: Git, CUDA Toolkit, Data Structure and Algorithms, Power BI, Tableau, Airflow, Docker, Flask, ChatGPT, Agile

Libraries & Frameworks: PyTorch, Tensorflow,Caffe, Scipy, Sklearn, Numpy, Pandas, Transformers, Seaborn, Keras, Matplotlib, OpenCV, SpeechRecognition, NLTK, Hugging Face, JAX

Generative AI: Vector Databases, Fine Tuning, Prompt Engineering, Quantization, PEFT, LoRA, GANs, VAEs

EXTRACURRICULARS

Social Work	Contributed 80+ hours for teaching in Educational Outreach programme of NSS, IITB [10/20-06/21]
	Volunteered in Blood Donation & Powai Lake Cleanup drive by UNICEF & Abhyuday, IITB [04/23,06/22]
	Volunteered in Food Distribution Drive organized by Robinhood Army(RHA) [10/22-Present]
Others	Organized a 14000 ft Trek for 20 members with IndiaHikes to Bhrigu Lake, Himachal Pradesh [07/21]
	Performed thematic hip-hop dance in Gyration, Inter-hostel Dance General Championship [02/22-03/22]